

torch.linalg.ldl_solve#

https://pytorch.org/docs/stable/generated/torch.linalg.ldl_solve.html

Description:

Computes the solution of a system of linear equations using the LDL factorization. LD and pivots are the compact representation of the LDL factorization and are expected to be computed by `torch.linalg.ldl_factor_ex()`. `hermitian` argument to this function should be the same as the corresponding arguments in `torch.linalg.ldl_factor_ex()`. Supports input of float, double, cfloat and cdouble dtypes. Also supports batches of matrices, and if A is a batch of matrices then the output has the same batch dimensions. This function is “experimental” and it may change in a future PyTorch release.

Function Signature

```
torch.linalg.ldl_solve(LD, pivots, B, *, hermitian=False, out=None) → Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

hermitian (bool, optional) – whether to consider the decomposed matrix to be Hermitian or symmetric. For real-valued matrices, this switch has no effect. Default: False. out (tuple, optional) – output tensor. B may be passed as out and the result is computed in-place on B. Ignored if None. Default: None.

Code Examples

```
>>> A = torch.randn(2, 3, 3)
>>> A = A @ A.mT # make symmetric
>>> LD, pivots, info = torch.linalg.ldl_factor_ex(A)
>>> B = torch.randn(2, 3, 4)
>>> X = torch.linalg.ldl_solve(LD, pivots, B)
>>> torch.linalg.norm(A @ X - B)
>>> tensor(0.0001)
```

Notes & Warnings

WARNING

Warning This function is “experimental” and it may change in a future PyTorch release.

Detailed Documentation

Documentation

Computes the solution of a system of linear equations using the LDL factorization.

LD and pivots are the compact representation of the LDL factorization and are expected to be computed by `torch.linalg.ldl_factor_ex()`. `hermitian` argument to this function should be the same as the corresponding arguments in `torch.linalg.ldl_factor_ex()`.

Supports input of float, double, cfloat and cdouble dtypes. Also supports batches of matrices, and if A is a batch of matrices then the output has the same batch dimensions.

This function is “experimental” and it may change in a future PyTorch release.

LD (Tensor) – the n times n matrix or the batch of such matrices of size $(*, n, n)$ where $*$ is one or more batch dimensions.

pivots (Tensor) – the pivots corresponding to the LDL factorization of LD.

B (Tensor) – right-hand side tensor of shape $(*, n, k)$.

`hermitian` (bool, optional) – whether to consider the decomposed matrix to be Hermitian or symmetric. For real-valued matrices, this switch has no effect. Default: False.

`out` (tuple, optional) – output tensor. B may be passed as `out` and the result is computed in-place on B. Ignored if None. Default: None.

